

DSA Assignment – Additional Tasks

Q1. Stack – Additional Task

Explain the time complexity and space complexity of each operation. Also discuss what happens when the stack size is fixed and the user attempts to insert more elements than its capacity.

Time and Space Complexity of Stack Operations

Operation	Time Complexity	Space Complexity
PUSH	$O(1)$	$O(1)$
POP	$O(1)$	$O(1)$
PEEK	$O(1)$	$O(1)$
DISPLAY	$O(n)$	$O(1)$

Fixed Stack Size: If the stack has a fixed capacity, it can store only a limited number of elements. When the stack is already full and the user tries to insert another element, a **Stack Overflow** condition occurs. The new element is not inserted and the program displays an overflow message.

Q2. Circular Queue – Additional Task

Compare the circular queue with a simple linear queue and explain: (1) why a circular queue provides better utilization of memory, (2) time complexity of ENQUEUE and DEQUEUE, (3) space complexity of the queue, and (4) the problem that occurs in a linear queue when REAR reaches the last index even though unused positions exist at the beginning.

1. Better Utilization of Memory

A circular queue provides better memory utilization because it reuses the empty positions created at the beginning of the array. After REAR reaches the last index, it can move back to the first index.

2. Time Complexity of ENQUEUE and DEQUEUE

ENQUEUE takes $O(1)$ time because an element is inserted directly at the rear. DEQUEUE also takes $O(1)$ time because the front position is moved to the next position.

3. Space Complexity of the Queue

The circular queue uses an array of fixed capacity. Therefore, the space used by the queue is $O(n)$, where n is the size or capacity of the array.

4. Problem in a Linear Queue

In a simple linear queue, when REAR reaches the last index, no more elements can be inserted from the rear. This can happen even when some positions at the beginning of the array are empty because of previous DEQUEUE operations. Therefore, available memory is not utilized efficiently.